BifocalGemma: Asymmetric Bidirectional Repository Encoding and Stationary Cross-Attention for Gemma 4 Developer Agents

Buana Daru Okta
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`https://github.com/daruokta/bifocalgemma`
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Abstract

Autoregressive decoder-only language models process software repositories under a fundamental structural bottleneck: causal attention masks prevent earlier repository tokens from attending to symbols, type definitions, or call sites that appear later in the linearized file stream, while prefix-concatenated prompts force the decoder to maintain massive key-value (KV) caches across tens of thousands of tokens during patch generation. We introduce **BifocalGemma**, an asymmetric encoder-decoder developer agent architecture built natively on the **Gemma 4** model family (`Gemma 4 E2B` and `Gemma 4 E4B`) [4]. BifocalGemma decouples repository comprehension from autoregressive patch synthesis by pairing a bidirectional Gemma 4 repository encoder ($N_{\text{enc}} \leq$ 32,768 tokens) with a causal Gemma 4 patch decoder ($N_{\text{dec}} \leq$ 8,192 tokens). To bridge these streams without destabilizing pretrained causal representations or breaking Gemma 4's native cross-layer KV-sharing topology, we introduce three architectural innovations: (1) a **SwiGLU Dual-Gate Coordinate Bridge** (`SwiGLUDualGateBridge`) with orthogonal/identity residual pathways, zero-initialized residual entry, and per-channel post-normalization calibration; (2) **Decoupled Stationary Full-Rank Cross-Attention** (`BifocalAttentionBlock`) that enforces rotational position invariance ($\Theta_{\text{cross}} = 0$), logarithmic **Critical Attention Length Scaling** $\beta(N_{\text{enc}})$, learnable per-head sharpness scaling σ_{cross}, and dual-gated head- and token-level fusion; and (3) a **Two-Stage Paired Training Curriculum** driven by **Chunked Differential Diff Cross-Entropy** ($\mathcal{L}_{\text{diff}}$). Across the 6-model controlled ablation matrix evaluated on `LongCodeArena` (`SWE-QA`) [2], `RepoBench-C-8k` [7], and `SWE-bench Lite` [6] under a strict Neutral Prompt Contract, **[PLACEHOLDER: Quantitative summary of primary benchmark outcomes across SWE-QA, RepoBench-C-8k, and SWE-bench Lite, alongside 32K memory and decoding throughput gains, to be populated upon completion of the active runs]**. All models are exported as standalone Hugging Face `PreTrainedModel` artifacts (`BifocalGemmaForCausalLM`) supporting native `AutoModelForCausalLM.from_pretrained(..., trust_remote_code=True)`.

1 Introduction

Repository-level software engineering requires two distinct computational modes: *global bidirectional comprehension* over multi-file dependency graphs and *precise local autoregressive synthesis* of unified diffs. Standard causal language models conflate both modes into a single left-to-right token stream. When a multi-file repository snapshot is linearized into a causal prefix $x_1, \ldots, x_N$, a declaration at token x_i cannot attend to downstream callers or interface implementations at token x_j $(j > i)$. Furthermore, during iterative reasoning and patch generation, every generated decoder token must attend over the entire N-token causal KV cache, incurring $\mathcal{O}(N_{\text{enc}}+N_{\text{dec}})$ memory bandwidth overhead at every decoding step.

Recent efforts to adapt causal backbones into bidirectional encoders [1, 3] demonstrate that unmasking self-attention unlocks strong bidirectional representations. However, grafting a bidirectional repository encoder onto a modern **Gemma 4** [4] decoder presents three acute architectural challenges:

1. **Subspace and Norm Mismatch Under Unmasked Self-Attention**: Removing the causal mask from a pretrained Gemma 4 backbone alters the spectral distribution and activation norms of the final hidden states, causing variance explosion if injected directly into a causal decoder.

2. **RoPE Coordinate Collision and Long-Context Entropy Dilution**: Standard Rotary Position Embeddings (RoPE) [9] assign sequential positions $t \in [0, N-1]$. Applying RoPE across decoupled encoder

and decoder streams creates artificial distance decay between a decoder token at step t_{dec} and relevant repository definitions at arbitrary file offsets $t_{\text{enc}} \gg t_{\text{dec}}$. Meanwhile, attending over $N_{\text{enc}} \in [8\text{K}, 32\text{K}]$ encoder tokens increases softmax entropy as $\mathcal{O}(\ln N_{\text{enc}})$, washing out sharp symbol lookups.

3. **Catastrophic Forgetting During Adapter Warmup**: Jointly updating decoder weights alongside randomly initialized cross-attention projections corrupts the decoder's fluent code generation capabilities before the cross-attention bridge has aligned its coordinate system.

Contributions. We present **BifocalGemma**, a modular encoder-decoder architecture engineered specifically for Gemma 4 developer agents:

- **Asymmetric Bidirectional-to-Causal Gemma 4 Topology**: We pair unmasked bidirectional Gemma 4 encoders (`E2B` at $d = 1536$, 35 layers; `E4B` at $d = 2560$, 42 layers) with causal Gemma 4 decoders while strictly isolating Per-Layer Embeddings (PLE, $d_{\text{ple}} = 256$) per stream and preserving Gemma 4's native cross-layer KV-sharing hierarchy (15 KV-owner + 20 KV-shared layers in `E2B`; 24 KV-owner + 18 KV-shared layers in `E4B`).

- **SwiGLU Dual-Gate Coordinate Bridge**: We design a residual manifold projector with width $d_{\text{mid}} = \max(d_{\text{enc}}, d_{\text{dec}})$, identity/orthogonal shortcut connections, zero-initialized down-projection, and learnable per-channel calibration γ_{calib} that seamlessly supports both symmetric (`E2B`→`E2B`, `E4B`→`E4B`) and heterogeneous (`E2B`→`E4B`, `E4B`→`E2B`) pairings.

- **Stationary Full-Rank Cross-Attention with Critical Length Scaling**: We formulate a RoPE-free ($\Theta_{\text{cross}} = 0$) cross-attention mechanism equipped with logarithmic length scaling $\beta(N_{\text{enc}})$, learnable per-head sharpness $\sigma_{\text{cross}} \in [0.1, 5.0]$, and a dual-gated injection mechanism initialized at 5% amplitude ($\tanh(g^{(0)}_{\text{head}}) = 0.05$).

- **Two-Stage Paired Curriculum & Differential Diff Objective**: We combine a frozen-decoder bridge alignment stage (Stage 1) and a conservative joint LoRA [5] SFT stage with 20% comprehension replay (Stage 2) under a memory-efficient Chunked Differential Diff Cross-Entropy loss ($\mathcal{L}_{\text{diff}}$).

- **Standalone Hugging Face Ecosystem Integration**: We package BifocalGemma as a self-contained Hugging Face `PreTrainedModel` (`BifocalGemmaForCausalLM`) with bundled architecture code and self-healing configuration assets.

2 Architecture and Methodology

2.1 Asymmetric Gemma 4 Backbone and Bidirectional Repository Encoder

Given a repository context X_{enc} of length $N_{\text{enc}} \leq 32{,}768$ and a task instruction q, we prepend a lightweight query prefix (`[QUERY] <instruction>\n`) to X_{enc} so that encoder self-attention layers condition repository representations on the target issue from layer 0. The encoder backbone is initialized from `Gemma 4` (`E2B` with $d_{\text{enc}} = 1536$, $L = 35$ layers, $H_Q = 8$, $H_{KV} = 1$; or `E4B` with $d_{\text{enc}} = 2560$, $L = 42$ layers, $H_Q = 8$, $H_{KV} = 2$).

To convert the causal Gemma 4 backbone into a true bidirectional repository encoder:

1. **Unmasked Bidirectional Attention**: We override the causal attention mask across all sliding-window and global full-attention layers (`use_bidirectional_attention="all"`), constructing a non-causal 4D padding mask $M_{\text{enc}} \in \{0, -\infty\}^{B\times 1\times 1\times N_{\text{enc}}}$ that permits every valid repository token i to attend to both preceding ($j < i$) and succeeding ($j > i$) tokens while strictly masking padding positions.

2. **Strict Stream-Isolated PLE**: Gemma 4 utilizes Per-Layer Embeddings (`hidden_size_per_layer_input` $= 256$) tied to surface token identities. We compute $E^{\text{enc}}_{\text{ple}}$ strictly from X_{enc} and $E^{\text{dec}}_{\text{ple}}$ strictly from X_{dec}, preventing cross-stream token identity leakage.

3. **Post-Encoding Padding Zeroing**: Output hidden states at padded positions are explicitly zeroed ($H_{\text{enc}} \leftarrow H_{\text{enc}} \odot \mathbb{I}[\text{mask}_{\text{enc}} = 1]$) before entering the coordinate bridge.

2.2 SwiGLU Dual-Gate Coordinate Bridge

To project bidirectional encoder representations $H_{\text{enc}} \in \mathbb{R}^{B\times N_{\text{enc}}\times d_{\text{enc}}}$ into the causal decoder's target manifold $Z_{\text{enc}} \in \mathbb{R}^{B\times N_{\text{enc}}\times d_{\text{dec}}}$ across arbitrary encoder-decoder width pairings, `SwiGLUDualGateBridge` applies a pre-normalized gated residual transformation [10, 8] with intermediate dimension $d_{\text{mid}} = \max(d_{\text{enc}}, d_{\text{dec}})$:

$$\tilde{H} = \text{RMSNorm}_{\text{pre}}(H_{\text{enc}}), \tag{1}$$

$$G = \text{SiLU}(\tilde{H}W_{\text{gate}}) \odot (\tilde{H}W_{\text{up}}), \quad W_{\text{gate}}, W_{\text{up}} \in \mathbb{R}^{d_{\text{enc}}\times d_{\text{mid}}}, \tag{2}$$

$$Z = W_{\text{res}}(\tilde{H}) + GW_{\text{down}}, \quad W_{\text{down}} \in \mathbb{R}^{d_{\text{mid}}\times d_{\text{dec}}}, \tag{3}$$

$$Z_{\text{enc}} = (\gamma_{\text{calib}} \odot \text{RMSNorm}_{\text{post}}(Z)) \odot \mathbb{I}[\text{mask}_{\text{enc}} = 1]. \tag{4}$$

Here, W_{res} is an identity mapping when $d_{\text{enc}} = d_{\text{dec}}$ (`E2B`→`E2B`, `E4B`→`E4B`) and an orthogonally initialized linear projection when $d_{\text{enc}} \neq d_{\text{dec}}$ (`E2B`→`E4B`, `E4B`→`E2B`).

The down-projection W_{down} is initialized near zero ($\mathcal{N}(0, (0.01/\sqrt{d_{\text{mid}}})^2)$) so the bridge begins as a clean linear/identity manifold transport, while $\gamma_{\text{calib}} \in \mathbb{R}^{d_{\text{dec}}}$ (initialized to $\mathbf{1}$) provides learnable per-channel scale calibration prior to cross-attention key/value projection.

2.3 Decoupled Stationary Full-Rank Cross-Attention

Every decoder layer wraps the native Gemma 4 attention module inside a `BifocalAttentionBlock` that executes causal self-attention and encoder-decoder cross-attention in parallel before merging their outputs through the shared output projection W_O.

Preservation of Gemma 4 Cross-Layer KV-Sharing. Gemma 4 employs a hybrid KV-sharing architecture in which only the first 15 of 35 layers in `E2B` (and 24 of 42 layers in `E4B`) compute independent KV projections (`is_kv_shared_layer = False`), while downstream shared layers reuse the KV states of their designated owner layer. BifocalGemma mirrors this exact invariant for cross-attention: only KV-owner layers instantiate dedicated BF16 full-rank projections $W_{K,\text{cross}}, W_{V,\text{cross}} \in \mathbb{R}^{d_{\text{dec}} \times (H_{KV} \cdot d_h)}$ and RMSNorms ($\text{RMSNorm}_{K,\text{cross}}, \text{RMSNorm}_{V,\text{cross}}$), warm-started from the pretrained self-attention W_K, W_V weights. During autoregressive generation, $(K_{\text{cross}}, V_{\text{cross}})$ are computed once during prefill, cached inside `BifocalDynamicCache`, and reused across both KV-shared layers and subsequent decoding tokens with $\mathcal{O}(1)$ recomputation cost.

Stationary Coordinate Invariance ($\Theta_{\text{cross}} = 0$). While causal self-attention applies standard RoPE rotations $\mathcal{R}_{\Theta,t}$ to Q_{self} and K_{self}, applying RoPE to cross-attention introduces spurious positional interference between the decoder's generation index and the encoder's file offset. Because Z_{enc} already encodes intra-file and inter-file structural topology from 35–42 layers of RoPE-equipped bidirectional self-attention, we enforce *Stationary Coordinate Invariance* ($\Theta_{\text{cross}} = 0$) on the cross-attention branch:

$$Q_{\text{cross}} = \text{RMSNorm}_{Q,\text{cross}}(H_{\text{dec}} W_Q), \tag{5}$$

$$K_{\text{cross}} = \text{RMSNorm}_{K,\text{cross}}(Z_{\text{enc}} W_{K,\text{cross}}), \tag{6}$$

$$V_{\text{cross}} = \text{RMSNorm}_{V,\text{cross}}(Z_{\text{enc}} W_{V,\text{cross}}). \tag{7}$$

Critical Attention Length Scaling and Learnable Sharpness. To prevent attention entropy explosion when querying up to $N_{\text{enc}} = 32{,}768$ repository tokens, we modulate Q_{cross} with both a deterministic logarithmic length normalizer $\beta(N_{\text{enc}})$ and a learnable per-head sharpness parameter $\sigma_{\text{cross}} \in \mathbb{R}^{H_Q \times 1 \times 1}$ (initialized to 1.0 and clamped to $[0.1, 5.0]$):

$$\beta(N_{\text{enc}}) = \max\left(1.0,\ 1.0 + 0.1 \ln\left(\frac{\max(N_{\text{enc}}, 1)}{2048}\right)\right), \tag{8}$$

$$\hat{Q}_{\text{cross}} = \frac{\text{clamp}(\sigma_{\text{cross}}, 0.1, 5.0)}{\beta(N_{\text{enc}})} Q_{\text{cross}}, \tag{9}$$

$$O_{\text{cross}} = \text{SDPA}\left(\hat{Q}_{\text{cross}}, K_{\text{cross}}, V_{\text{cross}}, M_{\text{enc}}\right). \tag{10}$$

Dual-Gated Head and Token Fusion. Rather than summing O_{self} and O_{cross} unconditionally, we combine a global per-head gate $g_{\text{head}} \in \mathbb{R}^{1 \times H_Q \times 1 \times 1}$ (initialized to $\text{atanh}(0.05)$ so $\tanh(g^{(0)}_{\text{head}}) = 0.05$) with a fine-grained token-conditional query gate parameterized by $w_{\text{gate}} \in \mathbb{R}^{1 \times H_Q \times 1 \times d_h}$ (initialized to $\mathbf{0}$ so $g^{(0)}_{\text{query}} = 1.0$):

$$g_{\text{query}} = 2\,\sigma\left(\frac{4}{\sqrt{d_h}} \langle Q_{\text{cross}}, w_{\text{gate}} \rangle\right) \in (0, 2), \tag{11}$$

$$g_{\text{eff}} = \tanh(g_{\text{head}}) \cdot g_{\text{query}}, \tag{12}$$

$$O_{\text{total}} = W_O\left(O_{\text{self}} + g_{\text{eff}} \odot O_{\text{cross}}\right). \tag{13}$$

At step 0 of training, cross-attention contributes a bounded 5% perturbation to the pre-projection attention representation, protecting the decoder's causal prior while preserving non-zero gradients into both the bridge and cross-attention projections.

2.4 Chunked Differential Diff Objective and Two-Stage Curriculum

Chunked Differential Diff Cross-Entropy ($\mathcal{L}_{\text{diff}}$). Standard uniform token-level cross-entropy underweights critical edit operations (`+` additions and `-` deletions) relative to boilerplate context lines inside unified diffs. We optimize *Chunked Differential Diff Cross-Entropy* ($\mathcal{L}_{\text{diff}}$), which parses target patches into line-level syntactic categories and assigns differential token weights: $w_{\text{add}} = 1.4$ for added lines (`+`), $w_{\text{del}} = 1.2$ for deleted lines (`-`), $w_{\text{hdr}} = 1.1$ for hunk headers (`@@ ... @@`, `--`, `+++`), and $w_{\text{ctx}} = 0.8$ for unchanged context lines. Weights are normalized per sequence to maintain unit mean ($\frac{1}{|S|}\sum_{t \in S} \tilde{w}_t = 1.0$). To minimize peak VRAM on $d_{\text{vocab}} = 262{,}144$, logits are computed in chunks of 128 tokens with Gemma 4 final logit softcapping (30.0):

$$\mathcal{L}_{\text{diff}} = \frac{1}{|S|} \sum_{t \in S} \tilde{w}_t \,\text{CE}\left(30.0 \tanh\left(\frac{h_t W_{\text{lm}}^\top}{30.0}\right), y_t\right). \tag{14}$$

Streaming pointer-generator copy loss is explicitly disabled ($\lambda_{\text{ptr}} = 0.0$) to prevent verbatim copy bias when synthesizing novel bug-fix tokens.

Two-Stage Paired Curriculum vs. Fair Baseline Curriculum.

- **Stage 1 (Frozen Decoder Bridge Alignment — First 50% of Steps)**: Both the encoder and decoder backbones are strictly frozen. Only `SwiGLUDualGateBridge` and the KV-owner `BifocalAttentionBlock` parameters are trained at $\text{LR}_{\text{bridge}} = \text{LR}_{\text{cross}} = 10^{-4}$ using comprehension and localization tasks.
- **Stage 2 (Joint LoRA SFT with Anti-Forgetting Replay — Final 50% of Steps)**: LoRA adapters ($r = 16, \alpha = 32$) are activated on both encoder and decoder with a conservative decoder learning rate $\text{LR}_{\text{lora}} = 2 \times 10^{-5}$, while bridge and cross-attention learning rates are halved to 5×10^{-5}. Stage 2 interleaves 80% unified diff synthesis (`SWE-smith`) with a 20% comprehension replay buffer (`SWE-QA` / `RepoBench`) to preserve long-context retrieval alignment.
- **Causal Baseline Curriculum**: Because causal baselines (`Baseline-E2B-IT`, `Baseline-E4B-IT`) have no randomly initialized bridge or cross-attention modules to warm up, they are trained under a continuous **Single-Stage Full Decoder LoRA SFT** across 100% of training steps on the exact same tokenized dataset and step budget.
- **Partitioned Weight Decay**: Weight decay ($\lambda_{\text{wd}} = 10^{-4}$) is applied strictly to 2D linear bridge matrices ($W_{\text{gate}}, W_{\text{up}}, W_{\text{down}}, W_{\text{res}}$) and set to 0.0 on all RMSNorm weights, calibration vectors γ_{calib}, cross-attention gates/scales, and LoRA adapters.

3 Experimental Setup

3.1 Controlled 6-Model Ablation Matrix and Benchmarks

Table 1 details our controlled 6-model experimental matrix spanning symmetric and asymmetric pairings of `Gemma 4 E2B` and `Gemma 4 E4B`. Models are trained on a deterministic multi-file repository corpus combining `SWE-smith` (unified diff bug repair), `LongCodeArena` (`SWE-QA` repository comprehension) [2], and `RepoBench-C-8k` [7]. We evaluate all models across four complementary dimensions:

1. **Long-Context Repository Comprehension (`SWE-QA`)**: Multi-hop architectural reasoning across contexts bucketed at 8K, 16K, and 32K tokens.
2. **Cross-File Code Retrieval (`RepoBench-C-8k`)**: Exact cross-file dependency resolution and completion accuracy.
3. **Automated Bug Localization & Patch Synthesis (`SWE-bench Lite`)**: Unified Diff Validity Rate, File/Hunk Localization Accuracy, and Exact Patch Resolution Rate [6].
4. **Hardware Efficiency Telemetry**: Peak inference VRAM (GB), prefill latency (ms), and decoding throughput (tokens/sec).

3.2 Evaluation Integrity Safeguards

Our benchmark harness enforces two strict invariants: (1) a **Neutral Prompt Contract** providing identical task instructions to both causal baselines and Bifocal-Gemma without architecture-specific prompt hints, and (2) a **256-Token Comprehension Generation Floor** (`max_new_tokens` ≥ 256) on all MCQ, QA, and retrieval tasks so analytical reasoning preambles are never truncated before emitting the final answer.

4 Empirical Results

[PLACEHOLDER — ACTIVE EXPERIMENTAL RUNS]: *All empirical metrics from earlier exploratory runs have been omitted. This section and Tables 2, 3, and 4 will be populated with final verified metrics upon completion of the active training and benchmark evaluation runs.*

4.1 Long-Context Repository Comprehension

Table 2 reports accuracy across `SWE-QA` (8K, 16K, 32K) and `RepoBench-C-8k`.

- **[PLACEHOLDER]**: Comparative analysis of repository comprehension accuracy (`SWE-QA` and `RepoBench-C-8k`) between Causal Baselines and BifocalGemma across 8K, 16K, and 32K token context buckets.
- **[PLACEHOLDER]**: Analysis of encoder-to-decoder scaling trade-offs (asymmetric `E2B`→`E4B` vs. `E4B`→`E2B`) on cross-file dependency retention.

4.2 Repository-Level Code Repair

Table 3 reports unified diff validity, localization accuracy, and patch resolution on `SWE-bench Lite`.

- **[PLACEHOLDER]**: Evaluation of unified diff synthesis quality, git patch syntactic validity, and file/hunk localization accuracy under the Two-Stage Paired Curriculum and Chunked Differential Diff Cross-Entropy ($\lambda_{\text{ptr}} = 0.0$).

4.3 Hardware Telemetry and Training Dynamics

Table 4 summarizes peak VRAM, prefill latency, and decoding throughput at 32K context length.

Table 1: Controlled 6-Model Ablation Matrix for BifocalGemma. All models use native Gemma 4 backbones with stream-isolated Per-Layer Embeddings ($d_{\text{ple}} = 256$) and preserve cross-layer KV-sharing (15/35 KV-owner layers in `E2B`; 24/42 KV-owner layers in `E4B`). Causal baselines use a single-stage full SFT curriculum, whereas BifocalGemma variants use the two-stage paired curriculum (Frozen Decoder Bridge Alignment → Joint LoRA SFT).

Model ID	Topology	Encoder Backbone	Decoder Backbone	Bridge ($d_{\text{enc}} \to d_{\text{mid}} \to d_{\text{dec}}$)	Context (N
`Baseline-E2B-IT`	Causal Only	None	`Gemma-4-E2B-IT` (35L)	—	0 / 8,1
`Baseline-E4B-IT`	Causal Only	None	`Gemma-4-E4B-IT` (42L)	—	0 / 8,1
`Bifocal-2B`	Symmetric	`Gemma-4-E2B` (35L)	`Gemma-4-E2B-IT` (35L)	$1536 \to 1536 \to 1536$ (Identity W_{res})	32,768 /
`Bifocal-4B-E2B`	Dec-Heavy	`Gemma-4-E2B` (35L)	`Gemma-4-E4B-IT` (42L)	$1536 \to 2560 \to 2560$ (Ortho W_{res})	32,768 /
`Bifocal-2B-E4B`	Enc-Heavy	`Gemma-4-E4B` (42L)	`Gemma-4-E2B-IT` (35L)	$2560 \to 2560 \to 1536$ (Ortho W_{res})	32,768 /
`Bifocal-4B`	Symmetric	`Gemma-4-E4B` (42L)	`Gemma-4-E4B-IT` (42L)	$2560 \to 2560 \to 2560$ (Identity W_{res})	32,768 /

Table 2: Long-Context Repository Comprehension and Cross-File Retrieval Results on `LongCodeArena` (`SWE-QA`) across context length buckets (8K, 16K, 32K) and `RepoBench-C-8k` under the Neutral Prompt Contract and 256-token generation floor. *Note: Placeholder cells (`[TBD]`) will be populated upon completion of the active training and evaluation runs.*

Model	`SWE-QA` @ 8K (%)	`SWE-QA` @ 16K (%)	`SWE-QA` @ 32K (%)	`SWE-QA` Overall (%)	`RepoBench-`
`Baseline-E2B-IT`	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD`
`Baseline-E4B-IT`	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD`
`Bifocal-2B` (`E2B` → `E2B`)	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD`
`Bifocal-4B-E2B` (`E2B` → `E4B`)	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD`
`Bifocal-2B-E4B` (`E4B` → `E2B`)	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD`
`Bifocal-4B` (`E4B` → `E4B`)	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD`

- **[PLACEHOLDER]**: Analysis of KV-cache memory efficiency from reusing Cross-KV states across Gemma 4 KV-shared layers and decoding throughput (tok/s) relative to causal baselines.
- **[PLACEHOLDER]**: Summary of Stage 1 (Bridge Alignment) and Stage 2 (Joint LoRA SFT) training and validation loss trajectories, alongside final learned distributions of $\tanh(g_{\text{head}})$ and sharpness σ_{cross}.

5 Limitations and Negative Findings

1. **Dual-Backbone Static Parameter Footprint**: Because BifocalGemma instantiates both a bidirectional encoder and a causal decoder, its static weight footprint in VRAM is roughly 2× that of a single-stream causal model of identical layer width (`Bifocal-2B` holds ∼ 4B total parameters across encoder and decoder), even though autoregressive decoding steps only traverse the decoder's causal KV cache and the precomputed 15/24 KV-owner cross-attention states.
2. **Sensitivity to Single-Stage Joint Warmup**: Skipping Stage 1 (Frozen Decoder Bridge Alignment) and training decoder LoRA adapters jointly from step 0 alongside unaligned cross-attention projections degrades unified diff formatting. Freezing the decoder during Stage 1 and initializing $\tanh(g_{\text{head}}^{(0)}) = 0.05$ is essential for stable convergence.
3. **Pointer-Generator Copy Bias on Code Repair**: Incorporating an explicit differentiable pointer-generator mixture ($\lambda_{\text{ptr}} > 0$) encourages the decoder to copy contiguous unchanged lines from the encoder context rather than emitting novel bug-fix tokens. Setting $\lambda_{\text{ptr}} = 0.0$ and relying on Chunked Differential Diff Cross-Entropy ($\mathcal{L}_{\text{diff}}$) resolves this failure mode.
4. **[PLACEHOLDER]**: Additional negative findings or boundary conditions observed in the final evaluation runs.

6 Conclusion

We presented **BifocalGemma**, an asymmetric encoder-decoder developer agent architecture built natively on the **Gemma 4** (`E2B` and `E4B`) model family. By combining unmasked bidirectional repository encoding ($N_{\text{enc}} \leq 32{,}768$), a norm-calibrated **SwiGLU Dual-Gate Coordinate Bridge**, **Decoupled Stationary Full-Rank Cross-Attention** ($\Theta_{\text{cross}} = 0$) with **Critical Attention Length Scaling** $\beta(N_{\text{enc}})$ and native Gemma 4

Table 3: Automated Bug Localization and Unified Diff Patch Synthesis on `SWE-bench Lite`. BifocalGemma variants are trained with Chunked Differential Diff Cross-Entropy ($\mathcal{L}_{\text{diff}}$, $\lambda_{\text{ptr}} = 0.0$) and a Two-Stage Paired Curriculum with 20% comprehension replay. *Note: Placeholder cells (`[TBD]`) will be populated upon completion of the active runs.*

Model	Valid Unified Diff Format (%)	File / Hunk Localization Acc. (%)	Exact Patch Resolutio
`Baseline-E2B-IT`	[TBD]	[TBD]	[TBD]
`Baseline-E4B-IT`	[TBD]	[TBD]	[TBD]
`Bifocal-2B` (E2B → E2B)	[TBD]	[TBD]	[TBD]
`Bifocal-4B-E2B` (E2B → E4B)	[TBD]	[TBD]	[TBD]
`Bifocal-2B-E4B` (E4B → E2B)	[TBD]	[TBD]	[TBD]
`Bifocal-4B` (E4B → E4B)	[TBD]	[TBD]	[TBD]

Table 4: Hardware Telemetry and Inference Efficiency at 32K Repository Context. Only KV-owner layers instantiate dedicated Cross-KV projections, reusing cached $(K_{\text{cross}}, V_{\text{cross}})$ across KV-shared layers during autoregressive generation. *Note: Placeholder cells (`[TBD]`) will be populated upon completion of the active runs.*

Model	Active Params (Enc + Dec)	KV-Owner Cross-Attn Layers	Peak VRAM @ 32K (GB)	Prefill Lat
`Baseline-E2B-IT`	2.0B	0	[TBD]	
`Baseline-E4B-IT`	4.0B	0	[TBD]	
`Bifocal-2B`	2.0B + 2.0B	15/35	[TBD]	
`Bifocal-4B-E2B`	2.0B + 4.0B	24/42	[TBD]	
`Bifocal-2B-E4B`	4.0B + 2.0B	15/35	[TBD]	
`Bifocal-4B`	4.0B + 4.0B	24/42	[TBD]	

cross-layer KV-sharing, and a **Two-Stage Paired Curriculum** driven by **Chunked Differential Diff Cross-Entropy**, BifocalGemma resolves the structural tension between global repository comprehension and local autoregressive patch generation.

[PLACEHOLDER — FINAL CONCLUSION SUMMARY]: *Empirical conclusions and practical deployment recommendations across symmetric and asymmetric Gemma 4 configurations will be inserted here upon completion of the active training and evaluation runs.*

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